Davronbek Malikov

LLM Engineer | RAG Engineer | Agentic AI Engineer | ML Engineer
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Summary

- LLM Engineer, RAG Engineer, Agentic AI Engineer, and ML Engineer with a Ph.D. in AI Convergence Engineering. I build LLM systems end to end: fine-tuning and inference serving, retrieval, agent orchestration, evaluation, and deployment on cloud and on-premise infrastructure.
- **LLM Engineering:** Supervised fine-tuning and parameter-efficient adaptation (LoRA/QLoRA), preference optimization (DPO), quantization (AWQ, GPTQ, GGUF), and local inference serving (vLLM, TGI, Ollama, llama.cpp).
- **RAG Engineering:** Standard RAG and GraphRAG pipelines with hybrid search (dense vectors + sparse BM25), cross-encoder reranking, hierarchical chunking, and vector databases (pgvector, Qdrant, Milvus).
- **Agentic AI Engineering:** Multi-agent architectures (Supervisor, Router, Network), plan-and-execute and self-correction loops, human-in-the-loop workflows, and tool use through MCP and function calling. Currently leading the AI engineering team at TripleH, shipping agents for legal research, financial analysis, meeting automation, and per-employee AI assistants.
- **ML Engineering:** Supervised and unsupervised machine learning on tabular data and Computer Vision, grounded in advanced mathematics, with 5 first-author publications in SCIE journals and international conferences.
- **Production, Evaluation & Security:** LLM-as-a-Judge evaluation (Ragas, DeepEval), input/output guardrails, prompt-injection defense, semantic caching, and token and cost tracking.

Education

Ph.D. in AI Convergence Engineering 2020–2025
Gyeongsang National University, Republic of Korea
Thesis: *Advancements in Soccer Analytics: Players' Goals and Salary Prediction Models with Machine Learning*
Supervisor: Jaeho Kim

Master of Informatics 2018–2020
Gyeongsang National University, Republic of Korea

Bachelor of Mathematics 2014–2018
Fergana State University, Uzbekistan

Experience

AI Engineer & Team Lead Dec 2025–Present
TripleH, Seoul, South Korea

- Leading the AI engineering team that designs, builds, and ships production-grade autonomous agent systems for enterprise clients.
- **Core stack:** open-weight local LLMs (Llama 4, Qwen3) served on-premise; post-training (SFT, LoRA/QLoRA) and alignment (DPO); RAG over vector databases (pgvector,

Qdrant) with hybrid search; agent orchestration with LangGraph and LangChain; tracing and evaluation with LangSmith.

- Architected and delivered:
  - **Multi-Agent Orchestration Platform** — Architected a central control hub that plans, routes, and monitors a network of specialized sub-agents: automated market reports, a financial-document RAG assistant, and a chatbot integrated with enterprise messenger apps.
  - **Legal Research Assistant** — Engineered a multi-LLM ensemble that sends each query to several models in parallel, aggregates their answers, and verifies every cited regional legal precedent before it reaches the user.
  - **Meeting Intelligence Agent** — Built on-premise speech-to-text with real-time transcription, automatic agenda-based minutes, and export to PDF, DOCX, Markdown, and JSON.
  - **On-Premise AI Platform** — Developed a deployment platform that runs every agent fully locally on open-weight models with no external API calls: zero data leakage and strict privacy and compliance for enterprise clients.
  - **Personal AI Assistants** — Built per-employee assistants that learn each user's tasks, communication style, and workflows to automate routine work.
- Driving end-to-end delivery, from research and prototyping to on-premise and cloud deployment.
- Setting engineering standards, running code reviews, and mentoring junior engineers on LLM integration and agent design.

Independent Researcher in Sports Analytics 2020–2025
Self-directed Research Projects

- Conduct independent research on the applications of LLMs and Deep Learning in soccer analytics.
- Develop models for performance evaluation, salary prediction, and tactical analysis using AI.
- Prepare and submit publications to SCIE and AI-focused journals.

Independent AI Educator (Machine Learning, Deep Learning, NLP, LLM) 2020–2025
Online and Private Instruction

- Design and deliver advanced courses in AI, Machine Learning, Deep Learning, and Large Language Models.
- Mentor students and professionals on building end-to-end AI systems and research-oriented projects.
- Focus on practical applications of AI in industry and academia.

Graduate Researcher 2022–2025
Big Data Systems Software Lab, Gyeongsang National University, South Korea

- Focused on ML applications in soccer analytics.
- **Publications:**
  - Malikov, D., et al. *Predicting Soccer Player Salaries with Both Traditional and Automated Machine Learning Approaches* (first author). SCIE: 17% increased accuracy of prediction model. [DOI]
  - Malikov, D., et al. *Beyond xG: A Dual Prediction Model for Analyzing Player Performance Through Expected and Actual Goals in European Soccer Leagues* (first author). SCIE: 12% improvement on player performance prediction. [DOI]
  - Malikov, D., et al. *Comparative Evaluation of Machine Learning Models for Predicting Soccer Injury Types* (first author). KSCIE: Introduced soccer players injury

prediction model using constructed features. [DOI]
  - Malikov, D., et al. *The Application of Machine Learning on the Injury Prediction of Soccer Players* (first author). ISE@APSEC: Novel approach for predicting players injury without GPS technology. [PDF]

Graduate Researcher 2018–2021
Distributed Systems Lab, Gyeongsang National University, South Korea

- Focused on applying advanced mathematical concepts to distributed systems.
- **Publications:**
  - On the construction and properties of lattice-group structure in Cartesian product spaces (first author). *Journal of Computer Science.* [DOI]

Mathematics Instructor (University Preparation, SAT & International Math Exams) 2014–2017
Fergana, Uzbekistan

- Taught advanced mathematics and analytical reasoning to high school graduates preparing for university entrance exams, the SAT, and other international mathematics exams.
- Developed customized teaching materials to strengthen students' problem-solving and logical thinking skills.

Projects

Touchline — AI Scouting Copilot for Football (Claude, LangGraph, RAG, LangSmith) [GitHub]

- Built an AI scouting assistant: a scout describes the player they need in plain language and gets a ranked shortlist and cited scouting reports.
- **Agent:** Claude orchestrated with LangGraph over 7 strict, typed tools (search, profile, similar players, compare, notes, shortlist); the model never writes SQL — Python validates arguments and builds the queries; 8-round tool cap per question.
- **Human-in-the-loop:** the agent pauses with a proposed shortlist (LangGraph `interrupt()`) and resumes from a persisted checkpoint once the scout approves.
- **RAG:** Qdrant hybrid search (dense bge-small + BM25) over scout notes and Wikipedia; reports use Claude citations, so every playing-style claim links to its source passage.
- **Data:** DuckDB warehouse of 50,149 players (Transfermarkt) joined with 297 StatsBomb matches → 18 per-90 metrics and percentiles for 594 players; 87.5% of players linked across sources.
- **Evaluation:** 120-question golden set in LangSmith with automatic checks (filter accuracy, precision/recall@10, numeric grounding, invented-player detection); 102 automated tests.
- Stack: Python, Claude API, LangGraph, LangSmith, Qdrant, DuckDB, FastAPI, Next.js, PostgreSQL.

TeachHelp — Post-Training an LLM into an Uzbek–English Teaching Assistant (SFT, DPO, RLHF) [GitHub] · **Live**

- Took a pretrained Qwen2.5 base model through the full pipeline — base → SFT → DPO → RLHF → deployment — and measured all 14 models on the same 68-question held-out test across 8 task types.
- **Spec-first workflow:** a written model spec defines language, answer style, tasks, limits, and the success test; training data is written to follow it and every model is judged against it.
- **SFT:** Qwen2.5-1.5B 15/68 → 44/68; three dataset versions (up to 1,583 examples) with test-leakage filtering; LoRA vs QLoRA under identical settings (4-bit cost only 2 points).

- **DPO from scratch:** sampled answers, rule-based chosen/rejected labels, frozen reference log-probabilities. v1 learned "shorter is better" (chosen shorter in 63% of pairs, 40/68); v2 fixed it with length-balanced pairs and an SFT loss term: **44/68**, all harmful and cheating requests refused (8/8) — the final model.
- **RLHF (InstructGPT recipe):** Bradley–Terry reward model + PPO with KL penalty and GAE, then RLOO; four judges (own 0.5B and 1.5B reward models, Skywork-Reward-V2, Gemini Flash-Lite as live judge, accuracy 0.72–0.90). Reward-hacking monitoring with a hidden rule score and a length-bias gate. Result 39–41/68: the policy was only as good as its judge, and DPO was simpler and better.
- **Deployment:** TeachHelp, a Claude-style chat in English and Uzbek on Vercel, served from a Kaggle GPU over a Cloudflare tunnel (∼2–3 s per answer); fully local version with all 14 models.
- Stack: Python, PyTorch, Hugging Face Transformers, PEFT, bitsandbytes, FastAPI, Kaggle, Vercel.

Portiq — Multi-Agent Job Hub (LangGraph, Kafka, A2A, pgvector) **Live**

- Built a multi-agent job hub for Uzbekistan, Korea and worldwide (incl. Telegram job channels) in Uzbek, Russian, Korean and English: job matching, skill-gap roadmaps and company-specific mock interviews.
- **Orchestration:** a LangGraph orchestrator plans each conversation (language → mode → country → only the missing questions) and routes tasks to 11 agents over Kafka; agents call each other over A2A (Agent Cards + JSON-RPC).
- **Quality gate:** a Judge agent checks every output and sends weak work back for a retry before the user sees it; interview answers scored on a rubric (knowledge, STAR structure, clarity, fit).
- **Job pipeline:** Remotive, Saramin, hh.uz and Telegram connectors → normaliser (so'm / won / $ salary parsing, dedup) → multilingual scam filter → Postgres + pgvector with an explainable match score (skills you have and miss, salary, visa).
- **Agents:** CV parsing (PDF/DOCX/TXT), 11-round mock interviews in 4 languages, 30/60/90-day roadmaps, company briefs from Wikipedia and Google News with every line sourced.
- **Engineering:** Claude, Gemini or Ollama with deterministic fallbacks; Jaeger tracing across agents; Docker Compose stack; unit tests, an end-to-end script and evals (scam precision/recall, rubric pairwise accuracy).
- Stack: Python, LangGraph, Kafka, FastAPI, PostgreSQL + pgvector, Redis, Next.js, Docker, Jaeger.

Soccer Player Salary Predictor — Industry-level ML Pipeline [GitHub] · **Live**

- Built an end-to-end supervised ML pipeline predicting professional soccer players' annual gross salaries from season-level performance statistics across Bundesliga, La Liga, and Serie A (8 seasons, 2014/15–2021/22).
- **Data leakage prevention:** Train/test split before any fitting; StandardScaler and OrdinalEncoder fitted on training data only; target salary never imputed at any stage.
- **Modeling tiers:** Baseline (Linear Regression, Ridge, Lasso, Decision Tree) → Improved (Random Forest, XGBoost, LightGBM, Gradient Boosting) → Ensemble (Stacking with Ridge meta-learner, Voting regressor).
- **Feature engineering:** 21 derived features without using the target — per-game rates (xG/game, assists/game), efficiency index $(goals+assists)/(xG+xA)$, contribution index, shot conversion, position simplification.
- **Hyperparameter tuning:** Optuna Bayesian optimisation (TPE sampler, 60 trials) over n_estimators, max_depth, learning_rate, subsample, and L1/L2 regularisation.
- **Explainability:** SHAP TreeExplainer — global beeswarm and bar charts, local waterfall

and force plots, engineered-feature impact report confirming FE added signal.
- **Deployment:** Gradio demo live on Render, kept active 24/7 via UptimeRobot (5-min ping). Full end-to-end daily logging throughout the pipeline.
- Linked to SCIE publication (Applied Sciences, 2025) — 17% improvement in prediction accuracy. [DOI]
- Stack: Python, XGBoost, LightGBM, Scikit-learn, SHAP, Optuna, Gradio, Render, UptimeRobot.

TacticalDNA — Unsupervised Player-Archetype Discovery [GitHub] · **Live**

- Companion project to the Salary Predictor: same Big-5-league dataset (Bundesliga, La Liga, Serie A), this time solving an unsupervised problem with zero salary/target leakage.
- Built an end-to-end clustering pipeline over 6,314 qualified player-seasons ($\geq$900 minutes) using only per-90 performance features — no salary, age, or team involved in the grouping.
- **Validated finding:** a blind stat line (25 goals, 15 assists, no name attached) was matched by the model to prime Messi's 2019/20 season at 98% cosine similarity — pure identity recovery from raw performance shape, with no player identity ever fed to the model.
- **Algorithm bake-off:** KMeans, Agglomerative (Ward), DBSCAN, and Gaussian Mixture compared on Silhouette, Davies-Bouldin, and Calinski-Harabasz scores; KMeans won outright.
- **Documented negative result:** DBSCAN collapses to a single dense cluster across every eps/min_samples tested — reported as a genuine empirical finding, not hidden.
- **Model selection trade-off:** chose $k = 6$ over the statistically optimal $k = 3$ for product interpretability — a deliberate statistical-optimum-vs-usefulness decision.
- **Archetype naming:** rule-based labeling from cluster centroids — Elite Complete Forward, Clinical Poacher, Deep-Lying Playmaker, Efficient Impact Player, Volume Forward, Defensive Anchor.
- **Deployment:** interactive Streamlit demo on Render; PCA(2) projection chosen over t-SNE/UMAP for stable, real-time inference on new player input.
- Stack: Python, Scikit-learn, Streamlit, Plotly, Render.

Palov — Halal Uzbek Food Platform for Korea **Live**

- Built a full-stack food ordering platform connecting the Uzbek community in South Korea with authentic halal cuisine.
- Implemented multi-country phone authentication (+82 KR / +998 UZ) with OTP verification and Telegram OAuth.
- **AI implementation:** Integrated a Retrieval-Augmented Generation (RAG) system over the menu database so the AI chatbot answers natural-language dish queries with real-time, context-aware recommendations.
- **Self-improvement loop:** User feedback and order patterns are collected and fed back into the recommendation model, allowing the AI to continuously improve suggestion quality over time.
- Stack: Next.js, LLM API, RAG, Vercel.

Hamyon — Personal Finance Manager (*"Pulingiz nazoratda"* — Your money under control) **Live**

- Built real-time expense tracking, budget monitoring, and category-based spending analysis.
- **AI implementation:** Integrated a RAG-powered AI chatbot over the user's financial transaction history, enabling natural-language queries (e.g. "How much did I spend on food this month?") and AI-generated budget insights.
- **Self-improvement loop:** The AI learns from user corrections and preference signals to continuously refine its categorization accuracy and financial advice quality.

- Stack: Next.js, Supabase, LLM API, RAG, Vercel.

Kotib — AI-Powered Personal Secretary (*"Aqlli shaxsiy kotib"*) **Live**

- Built the full AI assistant and chatbot core: natural-language task creation, scheduling, and intelligent note management.
- **AI implementation:** Designed a multi-step agent pipeline — user message → intent classification → action routing (create task / set reminder / generate note / query history) — powered by an LLM with Graph RAG over the user's personal knowledge base.
- **Self-improvement loop:** Tracks completed vs. missed tasks, analyses usage patterns, and autonomously adjusts proactive suggestions and reminder timing based on observed behaviour.
- Implemented persistent conversational memory maintaining full context across all sessions.
- Stack: Next.js, LLM API, Graph RAG, Vercel.

Tools & Frameworks

- **LLM Fine-Tuning & Adaptation:** Supervised Fine-Tuning (SFT), LoRA, QLoRA, Direct Preference Optimization (DPO), instruction tuning, continual pre-training, iterative feedback-driven training pipelines
- **Model Serving & Inference Optimization:** vLLM and TGI inference serving (PagedAttention), dynamic LoRA adapter hot-swapping, quantization (AWQ, GPTQ, GGUF), speculative decoding, local LLM deployment (Ollama, llama.cpp)
- **RAG & Vector Engineering:** Standard RAG, GraphRAG (knowledge graph integration), hybrid search (dense vector + sparse BM25), cross-encoder reranking (Cohere, BGE), hierarchical chunking and parent-child retrieval, vision-based layout parsing (LlamaParse, Unstructured.io), vector databases (pgvector, Qdrant, Milvus)
- **Agentic AI & Orchestration:** LangChain, LangGraph, multi-agent architectures (Supervisor, Router, Network), Mixture of Agents (MoA), plan-and-execute frameworks, self-correction and reflection loops, human-in-the-loop state persistence, tool use (Model Context Protocol, function calling)
- **Production, Evaluation & Security:** LLM-as-a-Judge evaluation (Ragas, DeepEval), LangSmith observability and tracing, input/output guardrails (NeMo Guardrails, Llama Guard), prompt injection and jailbreak defense, semantic caching (GPTCache), token tracking and cost optimization
- **Self-Improving AI Systems:**
  - *Feedback Collection:* Human feedback pipelines (thumbs up/down, ratings, corrections), preference data collection, user behaviour logging
  - *RLHF & Alignment:* Reinforcement Learning from Human Feedback (RLHF), reward model training, Proximal Policy Optimization (PPO), Direct Preference Optimization (DPO)
  - *Iterative Fine-Tuning:* Converting model outputs + human corrections into new training data; continual/online learning loops; automated retraining triggers
  - *Self-Reflection & Critique:* Chain-of-Thought self-correction, Constitutional AI (model critiques and revises its own outputs), LLM-as-Judge automated evaluation
  - *Knowledge & Memory Updates:* RAG knowledge base refresh from new interactions, Graph RAG graph expansion, persistent long-term conversational memory
  - *Prompt Optimization:* Behaviour-based prompt adjustment, automated prompt engineering (DSPy-style), A/B testing of prompt variants
  - *Synthetic Data Generation:* LLM-generated training examples, self-play data creation, data augmentation for low-resource tasks
  - *Active Learning:* Model flags low-confidence outputs for human review; uncertainty sampling to minimise labelling cost

- **Model Hubs & Testing:** Hugging Face (Transformers, PEFT, datasets, model hub), NVIDIA (CUDA, GPU-accelerated inference, NGC model registry, TensorRT)
- **ML / DL Frameworks:** Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy
- **Backend & DevOps:** FastAPI, Docker, GitHub Actions, SQL, Supabase, PostgreSQL (pgvector), Vercel, Render, UptimeRobot
- **Frontend & Web:** Next.js, React, HTML, CSS, JavaScript
- **Computer Vision:** YOLO, Roboflow, OpenCV
- **Research & Writing:** Overleaf, LaTeX

Competences

- **Agentic AI Engineering:** End-to-end design and deployment of multi-agent and Mixture-of-Agents systems for enterprise use cases.
- **LLM & RAG Engineering:** Taking LLM systems from fine-tuning and retrieval design through evaluation, guardrails, and production serving.
- **Team Leadership:** Leading engineering teams, conducting code reviews, and setting technical direction.
- **Research and Continuous Learning:** Eager to explore new AI tools and create models that contribute to science and industry.
- **Collaborative Development:** Works effectively in team environments using Git and GitHub.
- **Presentations:** Skilled at presenting complex ideas clearly to both technical and non-technical audiences.
- **Data Manipulation:** Proficient in SQL for handling and processing large datasets.

Languages

English (Fluent), Uzbek (Native), Turkish (Advanced), Russian (Intermediate), Korean (Conversational)